



WealthPilot

SME Loan Underwriting & Credit Research Assistant

Applied AI Professional Certification Program — IIT Hyderabad

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THE PROBLEM

Underwriting SME loans is slow, manual, and risky to get wrong

1

Manual review is slow

Credit-ops teams read applications and financial statements by hand, application by application.

2

Policy citations must be exact

A fabricated or misquoted policy citation isn't a UX bug here — it's a compliance failure.

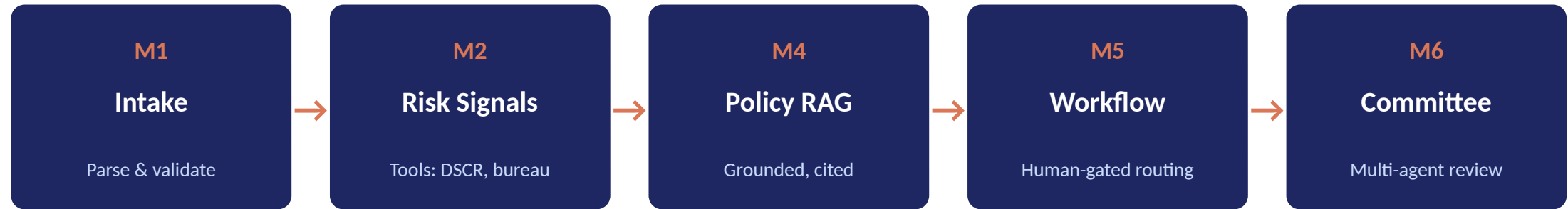
3

Bias risk is real and regulated

Similar financials with different applicant demographics must reach the same decision, provably.

WealthPilot automates the first pass — parse, verify, retrieve policy, decide — while keeping a human in the loop for every real decision.

Eight milestones, one pipeline



Runs underneath and alongside every stage above

M3 — Persistent Memory

Working / episodic / semantic / procedural, via Supermemory

M7 — Observability + Reliability

Tracing, retries, fallback, circuit breaker

M8 — Eval + Guardrails + API

Golden eval, safety layer, FastAPI deployment

From a raw application to real risk signals

M1

Structured Intake

- Provider-agnostic LLM client (LiteLLM) — swap providers in one line
- Raw record → validated Pydantic LoanApplication / FinancialSnapshot
- Repair loop retries on malformed output instead of failing outright
- Scored against each record's ground_truth for real accuracy, not just “did it validate”

10/10 records parsed successfully

M2

Tool-Enabled Agent

- Bounded tool-calling loop: DSCR calculator, interest calculator
- Credit-bureau and bank-statement lookups as real callable tools
- The model chooses which tool to call and when — not a hardcoded script
- Verified against a known eval case: computed DSCR matched by hand

DSCR, bureau & bank data — all real tool calls

Remembering context, answering only from real policy

M3

Persistent Memory

- Four memory kinds: working, episodic, semantic, procedural
- Backed by Supermemory — a real hosted memory store, not a mock
- Long conversations get compressed — lossy, so it's tested, not assumed
- Confirmed: a planted fact survives summarization, every run

Real Supermemory recall, semantic search verified

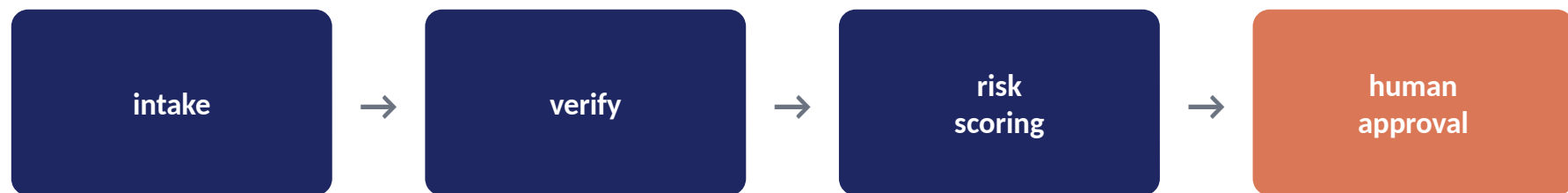
M4

Production RAG

- Local embeddings + BM25, fused with Reciprocal Rank Fusion
- Cross-encoder reranking before the model ever sees a chunk
- Every answer cites its source — fabricated policy = compliance failure here
- Prompt-injection defense: retrieved content is treated as untrusted data

Hybrid search + reranking, cited answers

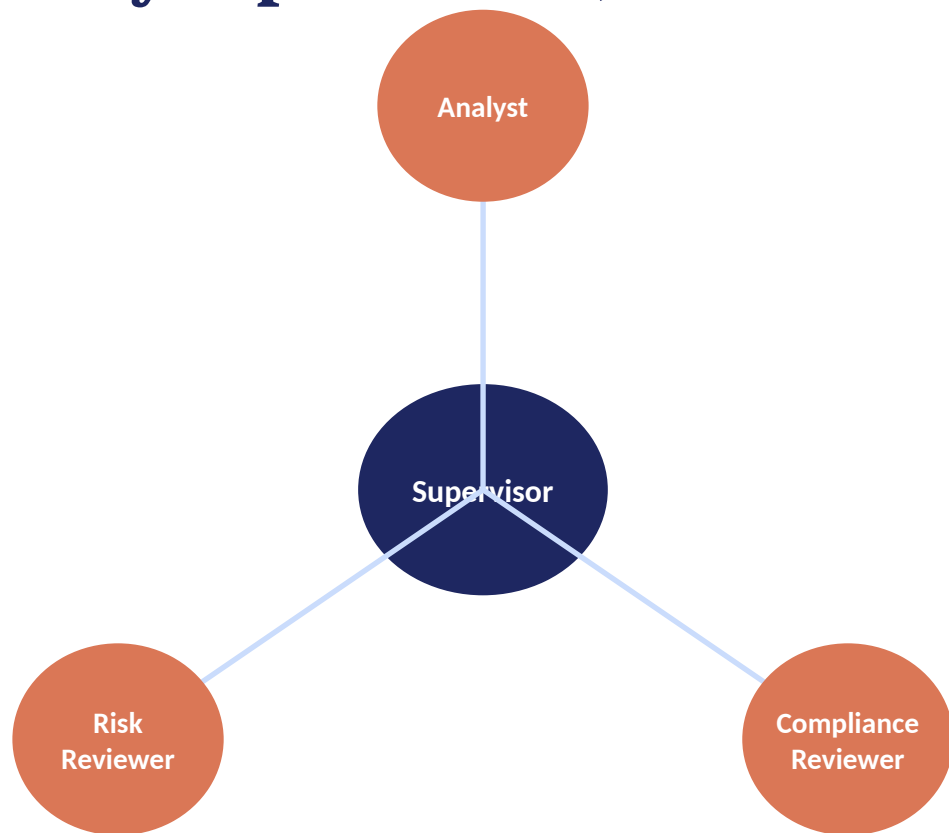
A real human-in-the-loop gate, not a simulated one



interrupt() pauses here — real pause, real resume

- Typed state with a control/audit field split — not a single overloaded dict
- A real `interrupt()` call pauses the graph and waits — not a node that just labels a decision
- A durable `SqliteSaver` checkpoint — verified surviving an actual process restart, not just an in-memory demo
- Bounded revision loop + escalation path — no unguarded loops

Analyst produces, two independent critics decide if it ships



- Enforced per-agent write scopes — a critic can't edit the memo it's checking
- Supervisor routes as a pure function, constrained to a `legal_routes()` guard
- Bounded revision loop + escalation — same discipline as M5
- Bureau/bank data served over a real MCP server, not a bare function call

Hardening the pipeline, then proving it

45% → 100%

Bureau-lookup success rate, before vs. after retry + fallback hardening (seeded fault injection)

3

Real failures the circuit breaker allowed before short-circuiting the rest — never hammered a dead dependency

20

golden eval items — deterministic checks + an LLM judge, scored honestly

- Observability: every node traced, tagged by run, with a real per-agent cost/latency dashboard
- Guardrails run independent of answer quality — schema validation, prompt-injection detection, protected-attribute scan
- 20/20 golden-set items passed guardrails clean; an injection attempt against the live API correctly returns HTTP 422
- FastAPI /health, /decision, /chat, deployed with a live public ngrok tunnel and verified from an external browser

RESULTS

What actually worked — and what didn't

M1 Intake

10/10 records parsed; missing-field accuracy checked against ground truth

M4 RAG

Grounded, cited answers; hybrid search + reranking measurably beats naive retrieval

M5 Workflow

Real interrupt → process restart → resume, verified, not simulated

M7 Reliability

Bureau lookup: 45% → 100% success under seeded fault injection

M8 Eval

19/20 golden-set items passed end-to-end; 20/20 guardrails clean; 3-judge score 1.0 faithfulness

Honest limitation:

The score started at 6/20 -- two real bugs found and fixed while verifying it, not a looser check. The eval generator was exploding citation-title strings into individual characters, and the citation checker didn't resolve the model's bracket-style citations. Both fixed at the source; the model's answers never changed.



Thank you

WealthPilot — all 8 milestones, built and independently verified.

github.com/abhi-1006/wealthpilot

Questions?